

Parity-factor Möbius capacity limit

RH research program

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Abstract

We isolate an exact constraint-graph theorem from a postcritically finite quadratic map. Its three-cell Markov partition has matrix $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, and the binary factor $A_{\{2\}}$ consists of words whose positive symbols lie in one parity class. For the Möbius prefix, the resulting adaptive capacity has an exact finite formula and satisfies $K_N^{(2)}/N \rightarrow 4/\pi^2$. The proof uses parity Mertens cancellation and odd/even squarefree densities. This is a reduced parity factor, not the four-state distance-two constraint of RH-366. The optimizer is chosen after reading the arithmetic prefix, so no canonical operator, prime trace, Riemann-zero model, Hilbert–Polya construction, or RH implication is claimed.

1 Scope and source boundary

The external source [1] proves that the restriction of $f_u(x) = 1 - ux^2$ at the real root

$$u_c^3 - 2u_c^2 + 2u_c - 2 = 0, \quad r = u_c - 1, \quad J = [-r, 1],$$

has the postcritical itinerary

$$0 \mapsto 1 \mapsto -r \mapsto r \mapsto r.$$

The partition $I_1 = [-r, 0]$, $I_2 = [0, r]$, $I_3 = [r, 1]$ has transition matrix

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \tag{1}$$

Labelling I_1 by 1 and the other intervals by 0 gives the factor $A_{\{2\}}$: all 1-positions lie in one parity class. The same source records the boundary-aware factor zeta function

$$\zeta_{f_{u_c}|_J}(z) = \zeta_{A_{\{2\}}}(z) = \frac{1+z}{1-2z^2}, \tag{2}$$

but (2) is used here only as a provenance anchor.

The distinction from RH-366 is essential. RH-366 permits positive symbols on both parity paths but forbids two positives at distance two; its capacity limit is still open. Here we study a distinct, explicitly defined reduced factor $A_{\{2\}}$, where all positive positions must use one parity class.

2 The finite parity-factor capacity

For $N \geq 1$, write $\mu_n = \mu(n)$ and

$$M_N = \sum_{n \leq N} \mu_n, \quad P_r(N) = \#\{n \leq N : n \equiv r \pmod{2}, \mu_n = 1\},$$

$$N_r(N) = \#\{n \leq N : n \equiv r \pmod{2}, \mu_n = -1\}, \quad r \in \{0, 1\}.$$

Define

$$K_N^{(2)} = \max_{\varepsilon \in (A_{\{2\}})_N} \left| \sum_{n=1}^N \mu_n \varepsilon_n \right|, \quad (3)$$

where $(A_{\{2\}})_N$ is the set of sign words for which the set $\{n : \varepsilon_n = +1\}$ is empty or contained in one residue class modulo 2.

Proposition 2.1 (Exact finite capacity). *For every N ,*

$$K_N^{(2)} = \max_{r \in \{0, 1\}} \max\{|-M_N + 2P_r(N)|, |-M_N - 2N_r(N)|\}. \quad (4)$$

Proof. Start with the all-negative word, whose score is $-M_N$. If positives are allowed only in parity class r , changing a position with $\mu_n = 1$ from -1 to $+1$ raises the score by 2, while changing a position with $\mu_n = -1$ lowers it by 2. Thus the largest score in that class is $-M_N + 2P_r(N)$, obtained by flipping every positive Möbius entry; the smallest is $-M_N - 2N_r(N)$, obtained by flipping every negative entry. Empty or partial flips cannot improve either extremum. Taking absolute values and then the two parity classes proves (4). $\square$

3 All-order limit

We use the standard consequences of the prime number theorem in arithmetic progressions and squarefree counting:

$$\frac{M_N}{N} \rightarrow 0, \quad \frac{P_1(N) + N_1(N)}{N} \rightarrow \frac{4}{\pi^2}, \quad \frac{P_0(N) + N_0(N)}{N} \rightarrow \frac{2}{\pi^2}. \quad (5)$$

The signed parity cancellation is an independent input: Davenport's fixed frequency estimate at $\theta = 1/2$ gives

$$A_N = \sum_{n \leq N} \mu(n)(-1)^n = o(N) \quad (\text{see [5]}).$$

Together with $M_N = o(N)$, this implies that the even and odd Möbius sums, $(M_N + A_N)/2$ and $(M_N - A_N)/2$, are both $o(N)$. The squarefree sieve gives the two densities in (5). Since $P_r - N_r = o(N)$ on each class, (5) gives

$$\frac{P_1(N)}{N}, \frac{N_1(N)}{N} \rightarrow \frac{2}{\pi^2}, \quad \frac{P_0(N)}{N}, \frac{N_0(N)}{N} \rightarrow \frac{1}{\pi^2}. \quad (6)$$

Theorem 3.1 (Parity-factor capacity limit). *For the factor $A_{\{2\}}$,*

$$\boxed{\lim_{N \rightarrow \infty} \frac{K_N^{(2)}}{N} = \frac{4}{\pi^2}}. \quad (7)$$

Proof. By (4) and (6), the two odd-parity candidates converge after division by N to $+4/\pi^2$ and $-4/\pi^2$, respectively. The two even-parity candidates converge to $+2/\pi^2$ and $-2/\pi^2$. The maximum of their absolute values therefore converges to $4/\pi^2$. $\square$

Remark 3.2 (What this does not prove). *The limit in (7) is not the unresolved RH-366 capacity for the distance-two four-state survivor. It is a capacity law for the distinct reduced factor $A_{\{2\}}$; the two languages are not comparable by set inclusion. Nor does (7) turn an adaptive sign word into a nonadaptive orbit, a canonical coupling, or a trace.*

4 Frozen executable audit

The package checks the integer Möbius sieve, (4) against exhaustive words through $N = 12$, and the source hashes. At $N = 2^{20}$ it records

$$K_N^{(2)} = 425095, \quad K_N^{(2)}/N = 0.40540218353271484.$$

This row is a reproduction diagnostic only. It is not a fit and is not used to establish (7).

5 Route and Gate ledger

Route A is **G0**: (1)–(7) give a source-locked PCF realization, an exact finite identity, and an all-order capacity theorem. Route B is **STOP_SCOPED**: the maximizing word reads the complete Möbius prefix, and the quantities are scalar adaptive sums. They are not a canonical dynamical determinant, a von Mangoldt prime-power trace, or a completed-zeta divisor. Gates A–E remain false/open. No Hilbert–Polya operator, Riemann-zero identification, or proof of RH follows.

References

- [1] Source package, *A postcritically finite quadratic realization of $W = 2$* , source commit 7fd3a3fdd5a6, 2026.
- [2] RH-366, *Möbius orthogonality, adaptive encoding, and Parry covariance*, repository release, 2026.
- [3] RH-367, *Boundary-aligned cyclic-Ulam structure and phase-local leakage*, repository release, 2026.
- [4] L. Mirsky, Arithmetical pattern problems relating to divisibility by r th powers, *Proc. London Math. Soc.* 50 (1949), 497–508.
- [5] H. Davenport, On some infinite series involving arithmetical functions (II), *Quart. J. Math.* 8 (1937), 313–320.